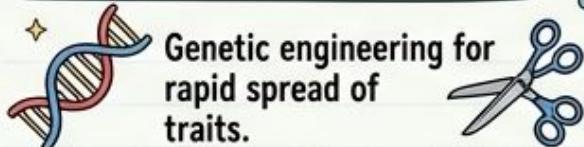


UPSC CSE PRELIMS 2026: CRUCIAL TOPICS OVERVIEW

ABHISHA AI
by Abhishek Pathak

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DNA session - 20/04/2026

(1. GENE DRIVE TECHNOLOGY & CRISPR-Cas9



Genetic engineering for rapid spread of traits.

- Uses CRISPR-Cas9 'molecular scissors'.
- Ensures >90% OFFSPRING inherit the gene.
- Enables specific genes to spread quickly in wild populations.
- APPLICATIONS:
 - Vector-borne Diseases: (e.g., modifying mosquitoes to prevent Malaria)
 - Invasive Species: (e.g., controlling agricultural pests)
 - Agriculture benefits (e.g., pesticide resistance)

3. BHARAT MARITIME INSURANCE POOL (BMIP)

- Domestic maritime insurance scheme.
- Backed by SOVEREIGN GUARANTEE.
- Primary for Indian flag vessels.
- Wide range of RISKS covered (e.g., war, environment).
- Initially run for 10 years.



2. ZERO-DAY VULNERABILITIES & AI IN CYBERSECURITY



Hidden security flaws in software/hardware.



- Unknown to the vendor.
- *Examples: Zoom flaws, Log4j.
- AI IN CYBERSECURITY:
 - Anthropic's Project Glasswing: Studying AI misuse (e.g., Claude mythos preview).
 - Uses of AI: Pattern recognition, threat detection.
- CONCERNS:
 - Risk of cartel-like control by a few large tech firms.
 - Centralized cartel structure in AI driven security ecosystem.
- *example project glasswing in AI driven cybersecurity.



centralized system

VS



decentralized ecosystem

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5. WEAKENING URBAN BREEZES

- Slower SEA BREEZE & LAND BREEZE.
- Impacts: Cool urban areas.
- Improves AIR QUALITY by flushing pollutants.



'normal'



'weakened' breeze

4. RBI PROPOSED KILL SWITCH



- For Card-based Payment Systems.

- Inspired by Singapore model.
- Enables disabling ALL DIGITAL TRANSACTIONS with one switch in case of fraud.

6. DIFFERENTIATING WELFARE AND DEVELOPMENT

WELFARE
(e.g., Subsidies, Cash transfers)



- Conceptual distinction.
- Immediate relief.
- Redistributive focus.

DEVELOPMENT
(e.g., Infrastructure, Education)



- Long-term structural process.
- Capacity building.

Trade-off



Balance

- Dangers of Populism/Partially Driven Transfers.
- Way Forward: Sustaining balanced growth.